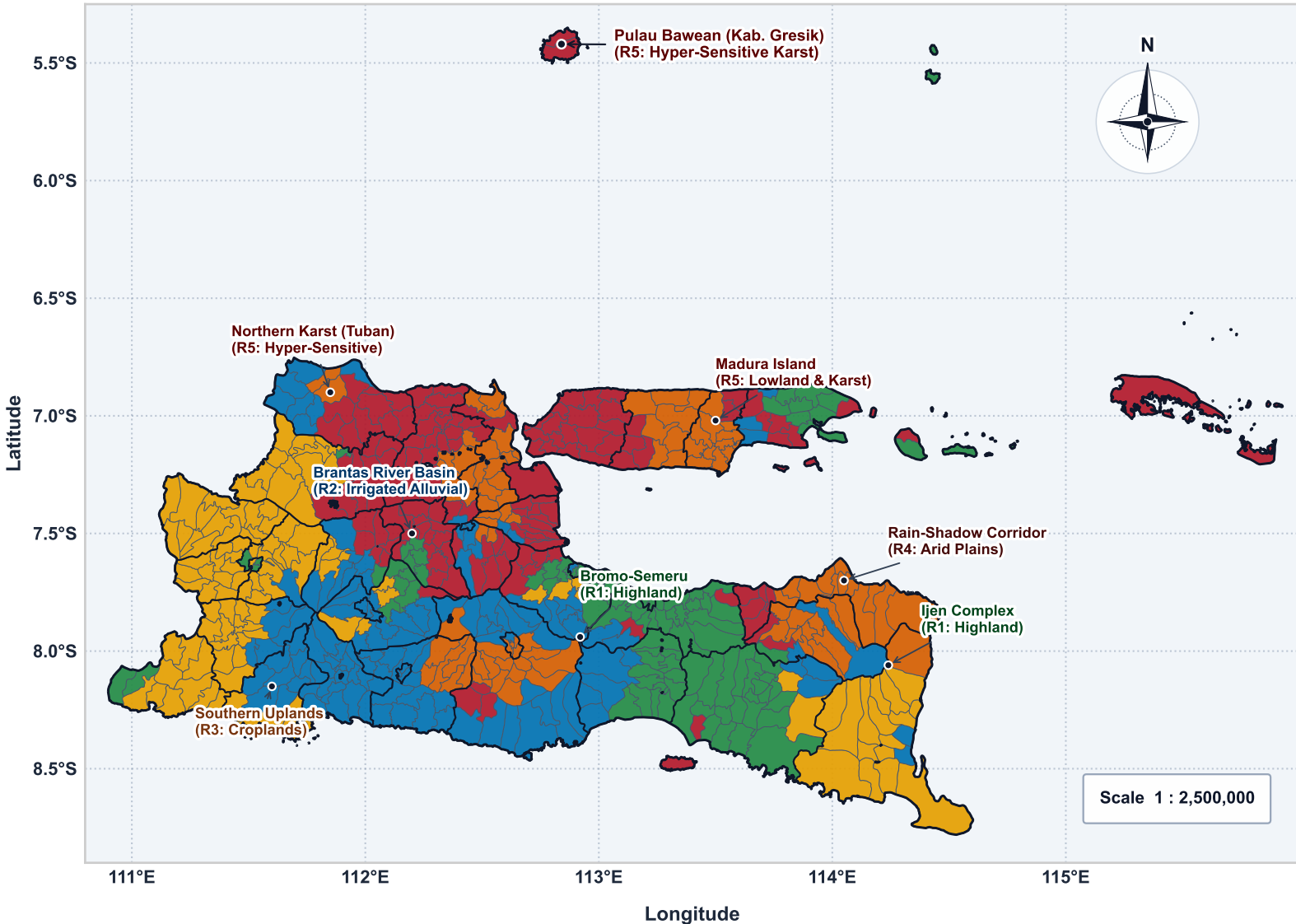


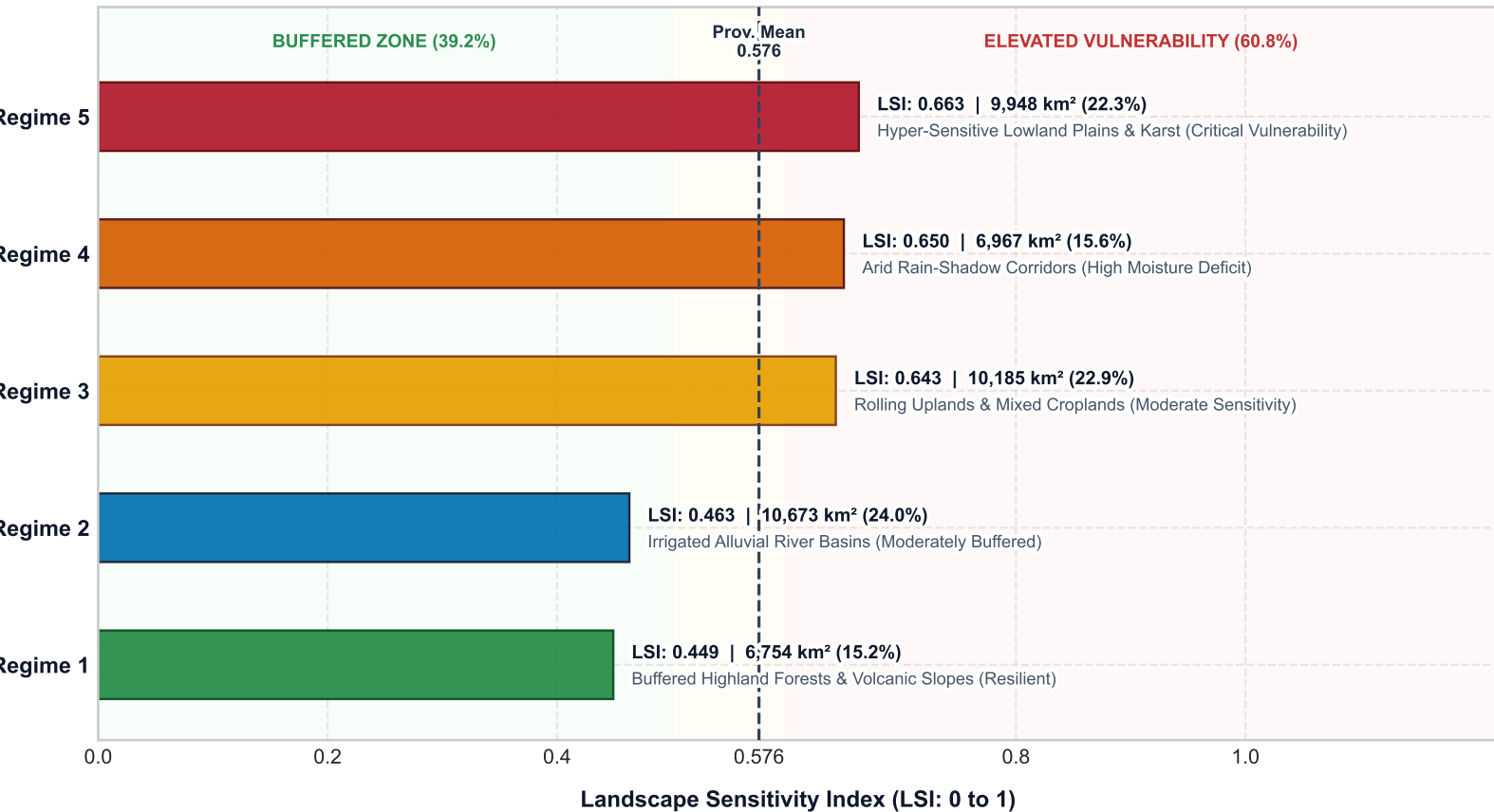
East Java Terrestrial Landscape Sensitivity & Response Regimes

Peak Dry Season (SON) Response Across Historical El Niño Episodes (2001–2025) | Weka k-Means (k = 5, n = 8,000 Spatial Samples, seed = 42) | Composite LSI (0 to 1, Prov. Mean = 0.576) | Total Landscape: 44,528 km²

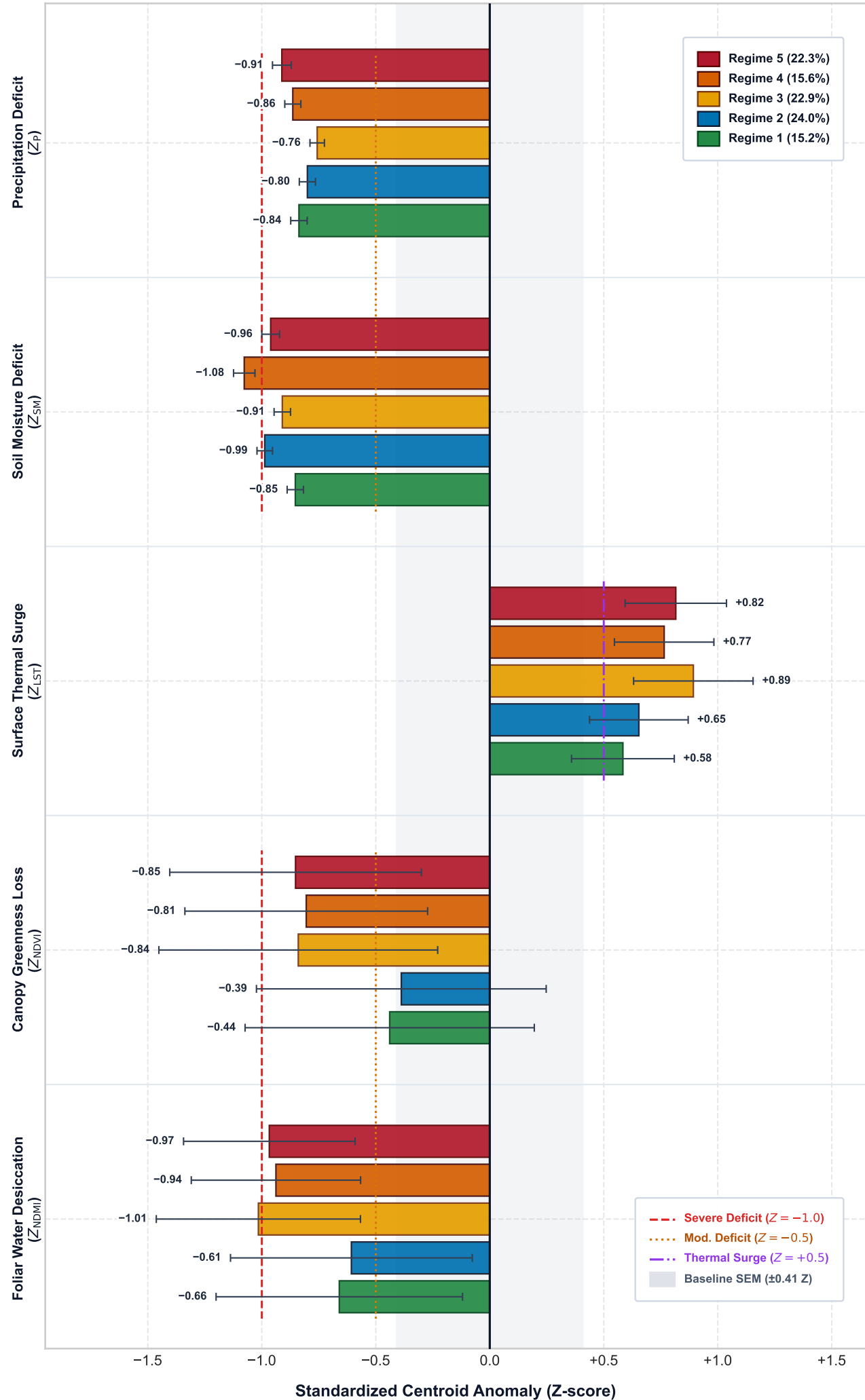
(a) Spatial Distribution of 5 Landscape Response Regimes Across East Java



(b) Hierarchical Landscape Sensitivity Ranking (LSI)



(c) Multi-Dimensional Landscape Response Signatures (LRS)



- ML Pipeline: Unsupervised k-Means (k=5, n=8,000 spatial samples, seed=42) on 5 standardized anomaly layers (1 km grid).
- LSI Formulation: $LSI = [(-Z_P) + (-Z_{SM}) + (Z_{LST}) + (-Z_{NDVI}) + (-Z_{NDMI}) - P_2] / (P_{98} - P_2)$, scaled to [0, 1].
- Within-Cluster Dispersion: Error bars denote ± 1 SD across 45,205 pixels (uniform macro-climate vs heterogeneous canopy).

- Biophysical Buffering: Volcanic orography & irrigation networks buffer R1–R2 (39.2% area); shallow soils compound R4–R5.
- Thermal-Moisture Coupling: R3 uplands exhibit peak thermal surge (+0.89 Z_{LST}), driving severe foliar desiccation (–1.02 Z_{NDMI}).
- Policy Implications: Priority targeted water allocation, reservoir quotas, and crop insurance for 27,100 km² in Regimes 3–5.